

CHAPTER 5

Molecular Basis of Inheritance

DNA

1. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Histones	(I)	Loosely packed chromatin
(B)	Nucleosome	(II)	Densely packed chromatin
(C)	Euchromatin	(III)	Positively charged basic proteins
(D)	Heterochromatin	(IV)	DNA wrapped around histone octamer

Choose the **correct** answer from the options given below:

- a. A-IV, B-III, C-II, D-I b. A-III, B-I, C-IV, D-II
c. A-II, B-III, C-IV, D-I d. A-III, B-IV, C-I, D-II

2. Given below are two statements: (2023)

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement I is incorrect but Statement II is true.
b. Both Statement I and Statement II are true.
c. Both Statement I and Statement II are false.
d. Statement I is correct but Statement II is false.

3. Match List-I with List-II: (2022 Re)

List-I		List-II	
(A)	Bacteriophage $\phi \times 174$	(I)	48502 base pairs
(B)	Bacteriophage Lambda	(II)	5386 nucleotides
(C)	<i>Escherichia coli</i>	(III)	3.3×10^9 Base pairs
(D)	Haploid content of human DNA	(IV)	4.6×10^6 base pairs

Choose the correct answer from the options given below:

- a. A-I, B-II, C-IV, D-III b. A-I, B-II, C-III, D-IV
c. A-II, B-IV, C-I, D-III d. A-II, B-I, C-IV, D-III

4. If A and C make 30% and 20% of DNA, respectively, what will be the percentage composition of T and G? (2022 Re)

- a. T : 20%, G : 20% b. T : 20%, G : 30%
c. T : 30%, G : 20% d. T : 30%, G : 30%

5. If the length of a DNA molecule is 1.1 metres, what will be the approximate number of base pairs? (2022)

- a. 6.6×10^6 bp b. 3.3×10^9 bp
c. 6.6×10^9 bp d. 3.3×10^6 bp

6. Read the following statements and choose the set of correct statements (2022)

- A. Euchromatin is loosely packed chromatin
B. Heterochromatin is transcriptionally active
C. Histone octamer is wrapped by negatively charged DNA in nucleosome
D. Histones are rich in lysine and arginine
E. A typical nucleosome contains 400 bp of DNA helix

Choose the correct answer from the options given below.

- a. A, C and E only b. B, D and E only
c. A, C and D only d. B and E only

7. Complete the flow chart on central dogma. (2021)

(A) $\xrightarrow{\text{DNA}}$ (B) $\xrightarrow{\text{mRNA}}$ (C) $\xrightarrow{\text{D}}$

- a. (A)-Translation; (B)-Replication;
(C)-Transcription; (D)- Transduction
b. (A)-Replication; (B)-Transcription;
(C)-Translation; (D)-Protein
c. (A)-Transduction; (B)-Translation;
(C)-Replication; (D)-Protein
d. (A)-Replication; (B)-Transcription
(C)-Transduction; (D)-Protein

8. If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it? (2021)

- a. T : 20; G : 20; C : 30 b. T : 30; G : 20; C : 20
c. T : 20; G : 25; C : 25 d. T : 20; G : 30; C : 20

9. Which one of the following statement about histones is **wrong**? (2021)

- a. The pH of histones is slightly acidic.
b. Histones are rich in amino acids - Lysine and Arginine.
c. Histones carry positive charge in the side chain.
d. Histones are organized to form a unit of 8 molecules.

10. Which of the following statements is correct? (2020)

- a. Adenine pairs with thymine through one H-bond
b. Adenine pairs with thymine through three H-bonds.
c. Adenine does not pair with thymine.
d. Adenine pairs with thymine through two H-bonds.

11. If the distance between two consecutive base pairs is 0.34 nm and the total number of base pairs of a DNA double helix in a typical mammalian cell is 6.6×10^9 bp, then the length of the DNA is approximately: (2020)
 a. 2.5 meters b. 2.2 meters
 c. 2.7 meters d. 2.0 meters
12. In the polynucleotide chain of DNA, a nitrogenous base is linked to the -OH of: (2020-Covid)
 a. 3'C pentose sugar b. 5'C pentose sugar
 c. 1'C pentose sugar d. 2'C pentose sugar
13. *E. coli* has only 4.6×10^6 base pairs and completes the process of replication within 18 minutes; then the average rate of polymerisation is approximately- (2020-Covid)
 a. 3000 base pairs/second b. 4000 base pairs/second
 c. 1000 base pairs/second d. 2000 base pairs/second
14. Purines found both in DNA and RNA are (2019)
 a. Adenine and thymine b. Adenine and guanine
 c. Guanine and cytosine d. Cytosine and thymine
15. The association of histone H_1 with a nucleosome indicates: (2017-Delhi)
 a. Transcription is occurring
 b. DNA replication is occurring
 c. The DNA is condensed into a chromatin fibre
 d. The DNA double helix is exposed
16. DNA fragments are: (2017-Delhi)
 a. Positively charged
 b. Negatively charged
 c. Neutral
 d. Either positively or negatively charged depending on their size
17. Identify the correct order of organisation of genetic material from largest to smallest: (2015 Re)
 a. Genome, chromosome, nucleotide, gene
 b. Genome, chromosome, gene, nucleotide
 c. Chromosome, genome, nucleotide, gene
 d. Chromosome, gene, genome, nucleotide
18. The diagram shows an important concept in the genetic implication of DNA. Fill in the blanks A to C: (2013)
- DNA $\xrightarrow{A}$ mRNA $\xrightarrow{B}$ Protein $\xrightarrow{\text{Proposed by C}}$
- a. A-transcription, B-extension, C-Rosalind Franklin
 b. A-transcription, B-replication, C-James Watson
 c. A-translation, B-transcription, C-Erwin Chargaff
 d. A-transcription, B-translation, C-Francis Crick
20. Given below are two statements: (2023)
Statement I: RNA mutates at a faster rate.
Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.
 In the light of the above statements, choose the correct answer from the options given below.
 a. Statement I is false but Statement II is true.
 b. Both Statement I and Statement II are true.
 c. Both Statement I and Statement II are false.
 d. Statement I is true but Statement II is false.
21. Which scientist conducted an experiment with ^{32}P and ^{35}S labelled phages for demonstrating that DNA is the genetic material? (2023-Manipur)
 a. James D. Watson and F.H.C. Crick
 b. A. D Hershey and M.J. Chase
 c. F. Griffith
 d. O.T. Avery, C.M. MacLeod and M. McCarty
22. Given below are two statements: (2023-Manipur)
Statement I: RNA being unstable, mutate at a faster rate.
Statement II: RNA can directly code for synthesis of proteins hence can easily express the characters.
 In the light of the above statements, choose the correct answer from the options given below:
 a. **Statement I** is correct but **Statement II** is incorrect
 b. **Statement I** is incorrect but **Statement II** is correct
 c. Both **Statement I** and **Statement II** are correct
 d. Both **Statement I** and **Statement II** are incorrect
23. With reference to Hershey and Chase experiments. Select the **correct** statements. (2023-Manipur)
 A. Viruses grown in the presence of radioactive phosphorus contained radioactive DNA.
 B. Viruses grown on radioactive sulphur contained radioactive proteins.
 C. Viruses grown on radioactive phosphorus contained radioactive protein.
 D. Viruses grown on radioactive sulphur contained radioactive DNA.
 E. Viruses grown on radioactive protein contained radioactive DNA.
 Choose the **most appropriate** answer from the options given below:
 a. (D) and (E) only b. (A) and (B) only
 c. (A) and (C) only d. (B) and (D) only
24. If DNA contained sulfur instead of phosphorus and proteins contained phosphorus instead of sulfur, what would have been the outcome of Hershey and Chase experiment? (2022 Re)
 a. Radioactive phosphorus in bacterial cells
 b. No radioactive sulfur in bacterial cells
 c. Both radioactive sulfur and phosphorus in bacterial cells
 d. Radioactive sulfur in bacterial cells
25. Ten *E. coli* with ^{15}N - dsDNA are incubated in medium containing ^{14}N nucleotide. After 60 minutes, how many *E. coli* cells will have DNA totally free from ^{15}N ? (2022)
 a. 80 cells b. 20 cells c. 40 cells d. 60 cells

The Search For Genetic Material And RNA World

19. Unequivocal proof that DNA is the genetic material was first proposed by (2023)
 a. Wilkins and Franklin
 b. Frederick Griffith
 c. Alfred Hershey and Martha Chase
 d. Avery, Macleod and McCarthy

26. The term 'Nuclein' for the genetic material was used by: (2020-Covid)
a. Miescher b. Chargaff c. Mendel d. Franklin
27. The experimental proof for semiconservative replication of DNA was first shown in a: (2018)
a. Fungus b. Bacterium c. Plant d. Virus
28. Select the correct match (2018)
a. Ribozyme Nucleic acid
b. $F_2 \times$ Recessive parent Dihybrid cross
c. T.H. Morgan Transduction
d. G. Mendel Transformation
29. The final proof for DNA as the genetic material came from the experiments of (2017-Delhi)
a. Griffith
b. Hershey and Chase
c. Avery, Mcleod and McCarty
d. Hargobind Khorana
30. A molecule that can act as a genetic material must fulfill the traits given below, except: (2016 - II)
a. It should be unstable structurally and chemically
b. It should provide the scope for slow changes that are required for evolution
c. It should be able to express itself in the form of 'Mendelian characters'
d. It should be able to generate its replica
31. Taylor conducted the experiment to prove semi- conservative mode of chromosome replication on: (2016 - II)
a. *Drosophila melanogaster* b. *E. coli*
c. *Vinca rosea* d. *Vicia faba*
32. Which of the following rRNA acts as structural RNA as well as ribozyme in bacteria? (2016 - II)
a. 23 S rRNA b. 5.8 S rRNA
c. 5 S rRNA d. 18 S rRNA
33. In sea urchin DNA, which is double stranded, 17% of the bases were shown to be cytosine. The percentages of the other three bases expected to be present in this DNA are: (2015)
a. G = 17%, A = 33%, T = 33%
b. G = 8.5 %, A = 50 %, T = 24.5 %
c. G = 34%, A = 24.5%, T = 24.5%
d. G = 17%, A = 16.5%, T = 32.5%
34. Which one of the following is not applicable to RNA? (2015 Re)
a. 5' phosphoryl and 3' hydroxyl ends
b. Heterocyclic nitrogenous bases
c. Chargaff's rule
d. Complementary base pairing
35. Transformation was discovered by: (2014)
a. Watson and Crick b. Messelson and Stahl
c. Hershey and Chase d. Griffith

Replication

36. Which of the following statement is correct regarding the process of replication in *E. coli*? (2024)
a. The DNA dependent DNA polymerase catalyses polymerization in $5' \rightarrow 3'$ as well as $3' \rightarrow 5'$ direction.
b. The DNA dependent DNA polymerase catalyses polymerization in $5' \rightarrow 3'$ direction.
c. The DNA dependent DNA polymerase catalyses polymerization in one direction that is $3' \rightarrow 5'$.
d. The DNA dependent RNA polymerase catalyses polymerization in one direction, that is $5' \rightarrow 3'$.
37. Given below are two statements: (2022 Re)
Statement I: DNA polymerases catalyses polymerisation only in one direction, that is $5' \rightarrow 3'$
Statement II: During replication of DNA, on one strand the replication is continuous while on the other strand it is discontinuous.
In the light of the above statements, choose the correct answer from the options given below
a. Statement I is incorrect but Statement II is correct
b. Both Statement I and Statement II are correct
c. Both Statement I and Statement II are incorrect
d. Statement I is correct but Statement II is incorrect
38. During DNA replication, Okazaki fragments are used to elongate (2017-Delhi)
a. The leading strand towards replication fork
b. The lagging strand towards replication fork
c. The leading strand away from replication fork
d. The lagging strand away from the replication fork
39. Select the correct option: (2014)

	Direction of RNA synthesis	Direction of reading of the template DNA strand
a.	$3' \rightarrow 5'$	$3' \rightarrow 5'$
b.	$5' \rightarrow 3'$	$3' \rightarrow 5'$
c.	$3' \rightarrow 5'$	$5' \rightarrow 3'$
d.	$5' \rightarrow 3'$	$5' \rightarrow 3'$

Transcription

40. Given below are two statements regarding RNA polymerase in prokaryotes. (2024 Re)
Statement-I: In prokaryotes, RNA polymerase is capable of catalysing the process of elongation during transcription.
Statement-II: RNA polymerase associate transiently with 'Rho' factor to initiate transcription.
In the light of the above statements, choose the correct answer from the options given below:
a. Statement-I is true but Statement-II is false
b. Statement-I is false but Statement-II is true
c. Both Statement-I and Statement-II are true
d. Both Statement-I and Statement-II are false

41. Given below are two statements: (2024 Re)

Statement-I: In eukaryotes there are three RNA polymerases in the nucleus in addition to the RNA polymerase found in the organelles.

Statement-II: All the three RNA polymerases in eukaryotic nucleus have different roles.

In the light of the above statements, choose the correct answer from the options given below:

- Statement-I is correct but Statement-II is incorrect
 - Statement-I is incorrect but Statement-II is correct
 - Both Statement-I and Statement-II are correct
 - Both Statement-I and Statement-II are incorrect
42. A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end; (2024)
- Inducer, Repressor, Structural gene
 - Promotor, Structural gene, Terminator
 - Repressor, Operator gene, Structural gene
 - Structural gene, Transposons, Operator gene
43. Which one is the correct product of DNA dependent RNA polymerase to the given template? (2024)
- 3'TACATGGCAAATATCCATTCA5'
- 5'AUGUACCGUUUUAUAGGGAAGU3'
 - 5'ATGTACCGTTTATAGGTAAGT3'
 - 5'AUGUACCGUUUUAUAGGUAAGU3'
 - 5'AUGUAAAGUUUUAUAGGUAAGU3'

44. Match List-I with List-II: (2024)

List-I		List-II	
(A)	RNA polymerase III	(I)	snRNPs
(B)	Termination of transcription	(II)	Promotor
(C)	Splicing of Exons	(III)	Rho factor
(D)	TATA box	(IV)	SnRNAs, tRNA

Choose the correct answer from the options given below:

- A-(III), B-(IV), C-(I), D-(II)
 - A-(IV), B-(III), C-(I), D-(II)
 - A-(II), B-(IV), C-(I), D-(III)
 - A-(III), B-(II), C-(IV), D-(I)
45. What is the role of RNA polymerase III in the process of transcription in Eukaryotes? (2023)
- Transcription of only snRNAs
 - Transcription of rRNAs (28S, 18S and 5.8S)
 - Transcription of tRNA, 5 srRNA and snRNA
 - Transcription of precursor of mRNA
46. Which one of the following is the sequence on corresponding coding strand, if the sequence on mRNA formed is as follow (2023)
- 5'AUCGAUCGAUCGAUCGAUCGAUCG 3'?
- 3' ATCGATCGATCGTCGATCGATCGATCG 5'
 - 5' UAGCUAGCUAGCUAGCUAGCUAGC UAGC 3'
 - 3' UAGCUAGCUAGCUAGCUAGCUAGCUAGC 5'
 - 5' ATCGATCGATCGATCGATCGATCGATCG 3'

47. Given below are two statements: (2023-Manipur)

Statement I: The process of copying genetic information from one strand of the DNA into RNA is termed as transcription.

Statement II: A transcription unit in DNA is defined primarily by the three regions in the DNA i.e., a promoter, the structural gene and a terminator.

In the light of the above statements, choose the correct answer from the options given below:

- Statement I** is true but **Statement II** is false
 - Statement I** is false but **Statement II** is true
 - Both **Statement I** and **Statement II** are true
 - Both **Statement I** and **Statement II** are false
48. What is the role of RNA polymerase III in the process of transcription in eukaryotes? (2021)
- Transcribes tRNA, 5s rRNA and sn RNA
 - Transcribes precursor of mRNA
 - Transcribes only snRNAs
 - Transcribes rRNAs (28S, 18S and 5.8S)
49. Identify the correct statement. (2021)
- RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.
 - The coding strand in transcription unit is copied to an mRNA.
 - Split gene arrangement is characteristic of prokaryotes.
 - In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.
50. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes? (2021)
- DNA dependent RNA polymerase
 - DNA Ligase
 - DNase
 - DNA dependent DNA polymerase
51. Name the enzyme that facilitates opening of DNA helix during transcription. (2020)
- DNA helicase
 - DNA polymerase
 - RNA polymerase
 - DNA ligase
52. AGGTATCGCAT is a sequence from the coding strand of a gene. What will be the corresponding sequence of the transcribed mRNA? (2018)
- AGGUAUCGCAU
 - UGGTUTCGCAT
 - ACCUAUGCGAU
 - UCCAUAAGCGUA
53. Spliceosomes are not found in cells of: (2017-Delhi)
- Plants
 - Fungi
 - Animals
 - Bacteria
54. Which of the following RNAs should be most abundant in animal cell? (2017-Delhi)
- r-RNA
 - t-RNA
 - m-RNA
 - mi-RNA
55. DNA-dependent RNA polymerase catalyses transcription on one strand of the DNA which is called the: (2016 - II)
- Alpha strand
 - Antistrand
 - Template strand
 - Coding strand

Genetic Code

56. In which disorder change of single base pair in the gene for beta globin chain results in change of glutamic acid to valine? (2023-Manipur)
- Thalassemia
 - Sickle cell anemia
 - Haemophilia
 - Phenylketonuria
57. Select the **correct** statements about sickle cell anaemia. (2023-Manipur)
- There is a change in gene for beta globin.
 - In the beta globin, there is valine in the place of Lysine.
 - It is an example of point mutation.
 - In the normal gene U is replaced by A.
- Choose the **correct** answer from the options given below:
- (B), (C) and (D) only
 - (B) and (D) only
 - (A), (B) and (D) only
 - (A) and (C) only
58. The salient features of genetic code are: (2023-Manipur)
- The code is palindromic
 - UGA act as initiator codon
 - The code is unambiguous and specific
 - The code is nearly universal
- Choose the **most appropriate** answer from the options given below:
- (A) and (D) only
 - (B) and (C) only
 - (A) and (B) only
 - (C) and (D) only
59. Against the codon 5' UAC 3', what would be the sequence of anticodon on tRNA? (2022 Re)
- 5' GUA 3'
 - 5' AUG 3'
 - 5' ATG 3'
 - 5' GTA 3'
60. **Statement I:** The codon 'AUG' codes for methionine and phenylalanine.
Statement II: 'AAA' and 'AAG' both codons code for the amino acid lysine.
 In the light of the above statements, choose the correct answer from the options given below. (2021)
- Both statement I and statement II are false
 - Statement I is correct but statement II is false
 - Statement I is incorrect but statement II is true
 - Both statement I and statement II are true
61. Under which of the following conditions will there be no change in the reading frame of following mRNA? (2019)
 5' AACAGCGGUGCUAUU 3'
- Insertion of G at 5th position
 - Deletion of G from 5th position
 - Insertion of A and G at 4th and 5th positions respectively
 - Deletion of GGU from 7th, 8th and 9th positions
62. Which of the following features of genetic code does allow bacteria to produce human insulin by recombinant DNA technology? (2019)
- Genetic code is not ambiguous
 - Genetic code is redundant
 - Genetic code is nearly universal
 - Genetic code is specific

63. If there are 999 bases in an RNA that codes for a protein with 333 amino acids, and the base at position 901 is deleted such that the length of the RNA becomes 998 bases, how many codons will be altered? (2017-Delhi)
- 1
 - 11
 - 33
 - 333
64. Which one of the following is the starter codon? (2016 - I)
- AUG
 - UGA
 - UAA
 - UAG

Translation

65. How many different proteins does the ribosome consist of? (2023)
- 20
 - 80
 - 60
 - 40
66. The process of translation of mRNA to proteins begins as soon as: (2022)
- The tRNA is activated and the larger subunit of ribosome encounters mRNA
 - The small subunit of ribosome encounters mRNA
 - The larger subunit of ribosome encounters mRNA
 - Both the subunits join together to bind with mRNA
67. Which of the following RNAs is **not** required for the synthesis of protein? (2021)
- tRNA
 - rRNA
 - siRNA
 - mRNA
68. The first phase of translation is: (2020)
- Recognition of DNA molecule
 - Aminoacylation of tRNA
 - Recognition of an anti-codon
 - Binding of mRNA to ribosome
69. Which one of the following is wrongly matched? (2014)
- Operon-Structural genes, operator and promoter
 - Transcription-Writing information from DNA to tRNA
 - Translation-Using information in mRNA to make protein
 - Repressor protein-Binds to operator to stop enzyme synthesis

Regulation of Gene Expression

70. Given below are two statements: (2024 Re)
- Statement I:** In the *lac* operon, the *z* gene codes for beta-galactosidase which is primarily responsible for the hydrolysis of lactose into galactose and glucose.
- Statement II:** In addition to lactose, glucose or galactose can also induce *lac* operon.
- In the light of the above statements, choose the correct answer from the options given below:
- Statement I is true but Statement II is false
 - Statement I is false but Statement II is true
 - Both Statement I and Statement II are true
 - Both Statement I and Statement II are false
71. The lactose present in the growth medium of bacteria is transported to the cell by the action of (2024)
- Permease
 - Polymerase
 - Beta-galactosidase
 - Acetylase

72. Match List-I with List-II (2024)

List-I		List-II	
(A)	Frederick Griffith	(I)	Genetic code
(B)	Francois Jacob & Jacque Monod	(II)	Semi-conservative mode of DNA replication
(C)	Har Gobind Khorana	(III)	Transformation
(D)	Meselson & Stahl	(IV)	Lac operon

Choose the correct answer from the options given below:

- a. A-(II), B-(III), C-(IV), D-(I)
 b. A-(IV), B-(I), C-(II), D-(III)
 c. A-(III), B-(II), C-(I), D-(IV)
 d. A-(III), B-(IV), C-(I), D-(II)

73. Match List-I with List-II. (2023)

List-I		List-II	
(A)	Gene 'a'	(I)	β -galactosidase
(B)	Gene 'y'	(II)	Transacetylase
(C)	Gene 'l'	(III)	Permease
(D)	Gene 'z'	(IV)	Repressor protein

Choose the correct answer from the option given below:

- a. A-(III), B-(I), C-(IV), D-(II)
 b. A-(II), B-(I), C-(IV), D-(III)
 c. A-(II), B-(III), C-(IV), D-(I)
 d. A-(III), B-(IV), C-(I), D-(II)

74. Name the component that binds to the operator region of an operon and prevents RNA polymerase from transcribing the operon. (2023-Manipur)

- a. Promotor
 b. Regulator protein
 c. Repressor protein
 d. Inducer

75. Which one of the following acts as an inducer for *lac* operon? (2023-Manipur)

- a. Sucrose b. Lactose c. Glucose d. Galactose

76. In *lac* operon, *z* gene codes for: (2022 Re)

- a. Transacetylase
 b. β -galactosidase
 c. Permease
 d. Repressor

77. Match List-I with List-II: (2022 Re)

List-I		List-II	
(A)	In <i>lac</i> operon <i>i</i> gene codes for	(I)	transacetylase
(B)	In <i>lac</i> operon <i>z</i> gene codes for	(II)	permease
(C)	In <i>lac</i> operon <i>y</i> gene codes for	(III)	β -galactosidase
(D)	In <i>lac</i> operon <i>a</i> gene codes for	(IV)	Repressor

Choose the correct answer from the options given below.

- a. A-(III), B-(I), C-(IV), D-(II)
 b. A-(III), B-(II), C-(I), D-(IV)
 c. A-(IV), B-(III), C-(II), D-(I)
 d. A-(I), B-(I), C-(III), D-(II)

78. In an *E. coli* strain *i* gene gets mutated and its product can not bind the inducer molecule. If growth medium is provided with lactose, what will be the outcome? (2022)

- a. RNA polymerase will bind the promoter region
 b. Only *z* gene will get transcribed
 c. *z*, *y*, *a* genes will be transcribed
 d. *z*, *y*, *a* genes will not be translated

79. Match the following genes of the *Lac* operon with their respective products : (2019)

(A)	<i>i</i> gene	(I)	β -galactosidase
(B)	<i>z</i> gene	(II)	Permease
(C)	<i>a</i> gene	(III)	Repressor
(D)	<i>y</i> gene	(IV)	Transacetylase

Select the correct option.

- a. A-(I), B-(III), C-(II), D-(IV)
 b. A-(III), B-(I), C-(II), D-(IV)
 c. A-(III), B-(I), C-(IV), D-(II)
 d. A-(III), B-(IV), C-(I), D-(II)

80. Select the correct match: (2018)

- a. Alec Jeffreys – *Streptococcus pneumoniae*
 b. Alfred Hershey and Martha Chase – TMV
 c. Matthew Meselson and F. Stahl – *Pisum sativum*
 d. Francois Jacob and Jacques Monod – *Lac* operon

81. All of the following are part of an operon except: (2018)

- a. An operator
 b. Structural genes
 c. An enhancer
 d. A promoter

82. The equivalent of a structural gene is: (2016 - II)

- a. Operon b. Recon c. Muton d. Cistron

83. Gene regulation governing lactose operon of *E. coli* that involves the *lac I* gene product is: (2015)

- a. Negative and repressible because repressor protein prevents transcription
 b. Feedback inhibition because excess of β -galactosidase can switch off transcription
 c. Positive and inducible because it can be induced lactose
 d. Negative and inducible because repressor protein prevents transcription

84. Which enzyme/s will be produced in a cell in which there is a nonsense mutation in the *lac Y* gene? (2013)

- a. Lactose permease and transacetylase
 b. β -galactosidase
 c. Lactose permease
 d. Transacetylase

Human Genome Project And DNA Fingerprinting

85. Expressed Sequence Tags (ESTs) refers to (2023)

- a. Certain important expressed genes.
 b. All genes that are expressed as RNA.
 c. All genes that are expressed as proteins.
 d. All genes whether expressed or unexpressed.

86. The last chromosome sequenced in Human Genome Project was: (2023-Manipur)
 a. Chromosome 6 b. Chromosome 1
 c. Chromosome 22 d. Chromosome 14
87. DNA polymorphism forms the basis of: (2022)
 a. Translation
 b. Genetic mapping
 c. DNA finger printing
 d. Both genetic mapping and DNA finger printing
88. If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as: (2022)
 a. Bioinformatics b. Sequence annotation
 c. Gene mapping d. Expressed sequence tags
89. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as: (2021)
 a. Repetitive DNA b. Single nucleotides
 c. Polymorphic DNA d. Satellite DNA
90. Which is the basis of genetic mapping of human genome as well as DNA finger printing? (2020-Covid)
 a. Single nucleotide polymorphism
 b. Polymorphism in hnRNA sequence
 c. Polymorphism in RNA sequence
 d. Polymorphism in DNA sequence
91. Expressed Sequence Tags (ESTs) refers to : (2019)
 a. Genes expressed as RNA b. Polypeptide expression
 c. DNA polymorphism d. Novel DNA sequences
92. Which of the following is not required for any of the techniques of DNA fingerprinting available at present? (2016 - I)
 a. Polymerase chain reaction
 b. Zinc finger analysis
 c. Restriction enzymes
 d. DNA-DNA hybridisation
93. Which of the following is required as inducer(s) for the expression of *Lac* operon? (2016 - I)
 a. Glucose b. Galactose
 c. Lactose d. Lactose and Galactose
94. Satellite DNA is important because it: (2015 Re)
 a. Shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which are heritable from parents to children.
 b. Does not code for proteins and is same in all members of the population
 c. Codes for enzymes needed for DNA replication
 d. Codes for proteins needed in cell cycle.
95. Commonly used vectors for human genome sequencing are: (2014)
 a. T/A Cloning Vectors b. T-DNA
 c. BAC and YAC d. Expression Vectors
96. An analysis of chromosomal DNA using the southern hybridisation technique does not use: (2014)
 a. PCR b. Electrophoresis
 c. Blotting d. Autoradiography

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (a) | 3. (d) | 4. (c) | 5. (b) | 6. (c) | 7. (b) | 8. (b) | 9. (a) | 10. (d) |
| 11. (b) | 12. (c) | 13. (d) | 14. (b) | 15. (c) | 16. (b) | 17. (b) | 18. (d) | 19. (c) | 20. (b) |
| 21. (b) | 22. (c) | 23. (b) | 24. (d) | 25. (d) | 26. (a) | 27. (b) | 28. (a) | 29. (b) | 30. (a) |
| 31. (d) | 32. (a) | 33. (a) | 34. (c) | 35. (d) | 36. (b) | 37. (b) | 38. (d) | 39. (b) | 40. (a) |
| 41. (c) | 42. (b) | 43. (c) | 44. (b) | 45. (c) | 46. (d) | 47. (c) | 48. (a) | 49. (a) | 50. (a) |
| 51. (c) | 52. (a) | 53. (d) | 54. (a) | 55. (c) | 56. (b) | 57. (d) | 58. (d) | 59. (a) | 60. (c) |
| 61. (d) | 62. (c) | 63. (c) | 64. (a) | 65. (b) | 66. (b) | 67. (c) | 68. (b) | 69. (b) | 70. (a) |
| 71. (a) | 72. (d) | 73. (c) | 74. (c) | 75. (b) | 76. (b) | 77. (c) | 78. (d) | 79. (c) | 80. (d) |
| 81. (c) | 82. (d) | 83. (d) | 84. (b) | 85. (b) | 86. (b) | 87. (d) | 88. (b) | 89. (a) | 90. (d) |
| 91. (a) | 92. (b) | 93. (c) | 94. (a) | 95. (c) | 96. (a) | | | | |